

TrustLens

AI-Powered Multimodal Trust Investigator

“Don’t trust the name. Investigate the evidence.”

Team PocketMind

Project Document

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1. Introduction

TrustLens is an AI-powered multimodal trust investigator designed to help users investigate internships, jobs, events, companies, recruiters, websites, and payment requests. It allows users to provide evidence through their phone and receive an evidence-based investigation with clear explanations.

Tagline: *“Don’t trust the name. Investigate the evidence.”*

2. Problem Statement

People frequently receive internship offers, job opportunities, event invitations, recruitment messages, and payment requests that may appear genuine. Scammers can use real company names, logos, fake recruiters, websites, offer letters, and payment links.

Users generally need to verify company registration, websites, email domains, recruiters, reviews, offer letters, and payment information separately. A company may be legally registered while a particular offer connected to it is still fake. TrustLens addresses the lack of a single phone-first workflow that connects and verifies these pieces of evidence.

3. TrustLens Solution

TrustLens provides a single multimodal investigation platform where users can capture or upload evidence and ask questions using voice. The system investigates the provided information and connects evidence from different sources. The core investigation process is:

See → Extract → Identify → Verify → Cross-check → Analyze → Explain → Act

TrustLens focuses not only on whether a company exists, but also on whether a specific opportunity, recruiter, payment request, or link is actually connected to that company.

4. Requirements

4.1 Functional Requirements

The system should support:

- Evidence capture and upload
- OCR and QR detection
- Entity extraction
- Company and opportunity verification
- Recruiter and payment verification
- Evidence cross-checking
- Risk assessment

- AI-generated explanations

4.2 Non-Functional Requirements

TrustLens should provide a fast, secure, mobile-friendly, explainable, and reliable investigation experience. Risk calculations should remain consistent, while AI-generated explanations should clearly show the evidence behind the result.

5. Workflow

From suspicion to confidence, in nine simple steps:

- Open TrustLens – launch the app and choose how to investigate (camera, upload, or voice).
- Capture evidence – scan posters, QR codes, business cards, or website content.
- Upload evidence – add offer letters, chat/email screenshots, PDFs, or documents.
- Ask with voice – speak a question such as “Is this internship genuine?”
- AI investigation – TrustLens analyzes the evidence, connects the data, and finds what matters.
- Get the complete report – a detailed investigation report with risk analysis and evidence breakdown.
- Take action – follow recommendations before paying, verifying, or registering.
- Multimodal input – use camera, upload, or voice, whichever works best.
- TrustLens complete – an informed decision backed by evidence.



6. Architecture

TrustLens follows a mobile-first architecture. The Android application handles user interaction, camera, uploads, voice, and suitable on-device processing. The backend manages investigation requests and communicates with AI, web investigation services, the risk engine, and the database.



6.1 Modules

TrustLens is divided into modules that handle different stages of the investigation:

Module	Purpose
User & Investigation	Starts and manages investigations
Camera & Upload	Captures and provides evidence
OCR & QR	Extracts text and QR information
Entity Extraction	Identifies companies, recruiters, offers, payments, etc.
Web Investigation	Retrieves current external evidence
Recruiter Verification	Checks recruiter identity and relationship
Payment Verification	Checks payment details and recipient

Module	Purpose
Cross-Check	Finds inconsistencies between evidence
Risk Assessment	Calculates the risk level
AI Explanation & Report	Explains findings and generates the final report

6.2 Multimodal Investigation

TrustLens accepts different forms of evidence so that an investigation is not limited to text alone. Users can point the camera at posters, logos, QR codes, payment requests, or websites. They can also upload PDFs, offer letters, screenshots, emails, and other documents.

Voice interaction allows users to ask questions such as **“Is this internship genuine?”** or **“Can I trust this registration link?”**

The system combines these different inputs to build a complete evidence picture.

6.3 AI / LLM Architecture

The AI architecture combines local processing with Gemini-based multimodal analysis. Gemini is used for multimodal understanding and for converting investigation evidence into an understandable explanation:

- Image / PDF / Voice input – collects evidence from camera, files, and voice.
- OCR / image / document processing – extracts text and processes images, documents and files.
- Entity & evidence extraction – identifies companies, opportunities, recruiters, payments, and key details.
- Gemini API – performs multimodal analysis and generates structured information.
- Structured investigation data – organizes extracted and analyzed information for further checks.
- Risk engine + AI explanation – calculates risk level using deterministic rules and explains findings with AI.
- Trust report – provides a complete investigation report with risk level, evidence, and recommendations.

6.4 Risk Engine

TrustLens uses a Python-based deterministic risk engine so that risk calculation follows consistent rules rather than depending only on an AI opinion. The system categorizes results into three levels:

Risk Level	Meaning
Low Risk	Evidence is broadly consistent

Risk Level	Meaning
Medium Risk	Some information needs additional verification
High Risk	Multiple significant inconsistencies or suspicious signals are detected

6.5 Database

TrustLens uses Supabase PostgreSQL for storing investigation-related data. This can include user information, investigation records, extracted entities, evidence metadata, risk results, and investigation history. Supabase Storage is used for uploaded files and investigation evidence.

6.6 APIs & Backend

The backend is developed using Python and FastAPI, providing a simple API layer between the Android application and the investigation services. The backend handles operations such as evidence upload, investigation requests, company verification, recruiter verification, payment verification, risk assessment, and trust report generation.

6.7 UI/UX

TrustLens follows a simple phone-first UI/UX so that investigation can happen directly in real-world situations.

7. Technology Stack

Layer	Technology	Use
Mobile App	Android + Kotlin + Jetpack Compose	Best fit for a phone-native product
Camera	CameraX	Easy Android camera integration
OCR	Google ML Kit	Fast local text recognition
QR	ML Kit Barcode Scanning	Easy QR/payment QR detection
Voice	Android SpeechRecognizer	Simple voice → text
AI	Gemini API	Multimodal images/PDFs + structured JSON
Backend	Python + FastAPI	Simple API development
Database	Supabase PostgreSQL	DB + Auth + Storage
Web Investigation	Search API + Website/Domain APIs	Get current evidence
Risk Engine	Python Deterministic Rules	Consistent risk calculation
AI Explanation	Gemini	Converts evidence into a report

Layer	Technology	Use
Deployment	Render / Railway	Simple FastAPI deployment
File Storage	Supabase Storage	File storage

8. Security & Privacy

TrustLens uses on-device processing where practical to improve privacy and responsiveness. Evidence should only be transmitted to backend or external services when required for investigation.

Uploaded evidence and investigation data should be protected through authenticated access and secure storage. API credentials and backend services should also be protected from unauthorized access.

9. Demo Scenario – iQOO Hackathon Poster

For the demonstration, the user points the TrustLens camera at an iQOO Hackathon poster. TrustLens investigates the event through the following steps:

- Camera – captures the poster image.
- OCR – extracts text from the image.
- QR detection – detects and reads the QR code.
- Event recognition – identifies the event and key details.
- Official source lookup – fetches information from official websites and official event sources.
- Cross-check – compares all available evidence.
- AI investigation – analyzes and verifies the data.
- Verified event information – provides the final report with details and recommendations.

“Can I trust this registration link?” The system investigates the link and explains the available evidence.

10. Expected Results

TrustLens is expected to identify relevant entities, verify available information, detect inconsistencies, and generate an understandable risk assessment.

The final output should provide the **risk level, supporting evidence, detected inconsistencies, and recommended next action**, allowing the user to make an informed decision.

11. Innovation & Uniqueness

TrustLens combines several investigation capabilities into one phone-first system. Its key differentiator is **evidence cross-checking**, where company, opportunity, recruiter, email/domain, payment, and website information are connected and compared.

Another important feature is **opportunity-level verification**. Instead of only checking whether a company exists, TrustLens checks whether the particular internship, job, event, or offer is actually connected to that company.

12. Limitations

TrustLens depends on the availability and accuracy of external information. Public information may be incomplete, unavailable, or outdated, and AI analysis may require additional verification.

The system therefore does not guarantee that an opportunity is genuine; it provides an evidence-based assessment using the information available to it.

13. Future Scope

Future development can include additional official data sources, advanced website analysis, multilingual voice interaction, expanded scam-pattern detection, continuous reputation monitoring, and more advanced document and communication analysis.

The evidence graph and investigation capabilities can also be expanded to support more complex investigations.

14. Conclusion

TrustLens provides a phone-first, multimodal, evidence-based investigation system for evaluating companies, opportunities, recruiters, events, websites, and payment requests.

By combining AI, on-device processing, web evidence, cross-checking, and deterministic risk analysis, TrustLens helps users investigate the evidence before they trust.

15. References

No.	Reference / Paper	Year	Relevance to TrustLens
1	Tahmasebi et al., “Multimodal Misinformation Detection using Large Vision-Language Models”	2024	Multimodal evidence & fact verification
2	Liu et al., “TextMonkey: An OCR-Free Large Multimodal Model for Understanding Document”	2024	Document understanding & information extraction
3	Greif et al., “Multimodal LLMs for OCR, OCR Post-Correction, and Named Entity Recognition”	2025	OCR & entity extraction

No.	Reference / Paper	Year	Relevance to TrustLens
4	"Detecting Online Recruitment Fraud Using Machine Learning"	2021	Recruitment & job fraud detection
5	"Fake Job Posting Detection using Machine Learning Algorithms"	2025	Fake job & recruiter detection
6	"Phishing Detection: A Recent Intelligent Machine Learning Comparison"	2017	Phishing & URL analysis
7	Duarte et al., "Machine Learning for Early Detection of Phishing URLs in Parked Domains"	2025	URL/domain verification
8	"Evolution of Phishing Detection: From Traditional Methods to Explainable AI"	2026	Explainable AI & risk analysis

16. Appendix

A. Sample Evidence

- Internship / Job Poster
- Offer Letter
- Recruiter Communication
- Payment QR Code
- Company Website / Registration Link

B. Evidence Relationship

Company → Opportunity → Recruiter → Email / Domain → Payment → Website

TrustLens cross-checks these entities to identify verified, mismatched, or unknown relationships and detect inconsistencies across the evidence.

C. Sample Risk Report

Risk Level: HIGH RISK

Key findings:

- Upfront registration fee requested
- Recruiter email domain does not match the official company domain
- Payment recipient does not clearly match the company
- Opportunity could not be independently confirmed through official sources

TrustLens Recommendation: Verify the opportunity through the company's official website or contact details before making any payment or sharing sensitive information.

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